

Important Instructions For This Test: Questions will generally become more difficult the further they are found in the test. Assume all matrix products and summations are defined. For a vector $\vec{v}$, v_i denotes the i -th element of $\vec{v}$. For a matrix M , M_{ij} denotes the element in the i -th row and j -th column of M . "NOTA" stands for "None of These Answers is correct."

~~~~~ Good Luck, and Have Fun! ~~~~~

1. Welcome to the FAMAT State Convention 2026 Matrices and Vectors test! The Matrices and Vectors test is incredibly unique among all FAMAT and MA $\Theta$  tests as it contains the highest concentration of university-level concepts despite being an Alpha topic! Most other FAMAT tests only cover the high school curriculum, so it takes a tremendous amount of bravery, knowledge, and skill to take the Matrices and Vectors test. For those of you who are, congratulations! As a previous Matrices and Vectors test taker myself, I, the test writer, have poured an incredible effort into creating what I believe is the greatest Matrices and Vectors test in MA $\Theta$  history! This test will challenge your Linear Algebra skills like never before, so to all who have read this paragraph, I wish you the best of luck!

Since this is only the beginning of the test, we'll start off with an easy question. Compute  $\left| \begin{bmatrix} 2 & 0 \\ 2 & 6 \end{bmatrix} - \begin{bmatrix} 0 & -2 \\ 2 & -6 \end{bmatrix} \right|$ .

- (A) 0                      (B) 8                      (C) 16                      (D) 24                      (E) NOTA

2. Let  $\vec{u} = \langle 2, 2, -6 \rangle$ ,  $\vec{v} = \langle 7, 7, 7 \rangle$ , and  $\vec{w} = \langle -1, -2, 3 \rangle$ . Evaluate  $\vec{u} - \vec{w} + 2\vec{v}$ .

- (A)  $\langle 17, 18, 5 \rangle$                       (B)  $\langle -7, -9, -7 \rangle$                       (C)  $\langle 15, 14, 11 \rangle$                       (D)  $\langle 10, 11, -2 \rangle$                       (E) NOTA

3. Find  $\left( \begin{bmatrix} 6 & 5 & 2 \\ 10 & -10 & 10 \\ 7 & 7 & 7 \end{bmatrix} + \begin{bmatrix} -1 & 13 & 0 \\ -5 & 6 & 3 \\ 7 & 7 & 7 \end{bmatrix} + \begin{bmatrix} -2 & 0 & -8 \\ -5 & 2 & 3 \\ 7 & 7 & 7 \end{bmatrix} \right)_{23}$ .

- (A) -2                      (B) 16                      (C) 18                      (D) 21                      (E) NOTA

4. Let  $\gamma$  be the measure of the smaller of the two angles formed by  $\langle 6, -6 \rangle$  and  $\langle -1, 7 \rangle$ . Find  $\sin \gamma$ .

- (A)  $-\frac{4}{5}$                       (B)  $-\frac{3}{5}$                       (C)  $\frac{3}{5}$                       (D)  $\frac{4}{5}$                       (E) NOTA

5. Which of the following is equal to  $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^{2026}$ ?

- (A)  $\begin{bmatrix} 1 & 2^{2025} \\ 0 & 1 \end{bmatrix}$                       (B)  $\begin{bmatrix} 1 & 2^{2026} \\ 0 & 1 \end{bmatrix}$                       (C)  $\begin{bmatrix} 1 & 2025 \\ 0 & 1 \end{bmatrix}$                       (D)  $\begin{bmatrix} 1 & 2026 \\ 0 & 1 \end{bmatrix}$                       (E) NOTA

6. Which of the following matrices is nilpotent?

- (A)  $\begin{bmatrix} 5 & 5 \\ 1 & 1 \end{bmatrix}$                       (B)  $\begin{bmatrix} -2 & -6 \\ 1 & 3 \end{bmatrix}$                       (C)  $\begin{bmatrix} 3 & -9 \\ -1 & 3 \end{bmatrix}$                       (D)  $\begin{bmatrix} 3 & 9 \\ -1 & -3 \end{bmatrix}$                       (E) NOTA

7. Which of the following matrices is the reduced row-echelon form of  $\begin{bmatrix} 7 & -3 & 1 \\ 1 & 7 & -3 \\ -3 & 1 & 7 \end{bmatrix}$ ?

- (A)  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$       (B)  $\begin{bmatrix} 1 & -7 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$       (C)  $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{bmatrix}$       (D)  $\begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & -5 \\ 0 & 0 & 0 \end{bmatrix}$       (E) NOTA

8. Find the volume of the tetrahedron formed by the points  $\langle 5, 6, 5 \rangle$ ,  $\langle 2, -3, 8 \rangle$ ,  $\langle 8, 9, 10 \rangle$ , and  $\langle 7, 7, 7 \rangle$ .

- (A) 7      (B) 8      (C) 9      (D) 10      (E) NOTA

9. Consider

$$\mathcal{G} = \begin{bmatrix} 4 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -3 & 7 \\ -5 & 12 \end{bmatrix} \begin{bmatrix} 9 & -8 \\ -6 & 7 \end{bmatrix} \begin{bmatrix} -15 & 9 \\ 12 & -7 \end{bmatrix}$$

Which of the following is equal to the determinant of  $\mathcal{G}$ ?

- (A) -45      (B) 45      (C) 90      (D) 135      (E) NOTA

**Please Use the Following Information to Answer Questions 10 to 12:** Let  $\mathcal{G} = \begin{bmatrix} -2 & 3 & -2 \\ 1 & 0 & 5 \\ 0 & -1 & -4 \end{bmatrix}$ .

10. Compute  $\det \mathcal{G}$ .

- (A) -20      (B) -12      (C) 4      (D) 24      (E) NOTA

11. Which of the following is equal to  $\mathcal{G}^{-1}$ ?

- (A)  $\begin{bmatrix} \frac{5}{4} & \frac{7}{2} & \frac{15}{4} \\ 1 & 2 & 2 \\ -\frac{1}{4} & -\frac{1}{2} & -\frac{3}{4} \end{bmatrix}$       (C)  $\begin{bmatrix} -1 & -1 & 0 \\ \frac{1}{2} & 2 & \frac{1}{2} \\ \frac{1}{6} & -\frac{5}{12} & -\frac{7}{12} \end{bmatrix}$       (E) NOTA
- (B)  $\begin{bmatrix} \frac{5}{12} & \frac{7}{12} & \frac{17}{12} \\ \frac{1}{4} & \frac{1}{3} & -\frac{1}{2} \\ -\frac{7}{12} & \frac{2}{3} & -\frac{3}{4} \end{bmatrix}$       (D)  $\begin{bmatrix} -\frac{1}{5} & \frac{1}{20} & \frac{2}{5} \\ \frac{1}{5} & \frac{3}{20} & \frac{3}{10} \\ 0 & -\frac{2}{5} & \frac{1}{2} \end{bmatrix}$

12. Let  $M_C$  denote the cofactor matrix of  $M$ . Compute  $\det(\mathcal{G}_C)$ .

- (A) 16      (B) 64      (C) 144      (D) 256      (E) NOTA

**Please Use the Following Information to Answer Questions 13 to 14:** Let  $\vec{u} = \langle 2, -1, 3 \rangle$  and  $\vec{v} = \langle -2, 5, 1 \rangle$ . Additionally, define  $\vec{w} = \text{proj}_{\vec{v}}(\vec{u})$ .

13. Find  $\vec{u} \times \vec{v}$ .

- (A)  $\langle -4, -5, -3 \rangle$       (B)  $\langle -16, 8, 8 \rangle$       (C)  $\langle 16, 8, 8 \rangle$       (D)  $\langle -16, -8, 8 \rangle$       (E) NOTA

14. Compute  $\|(\vec{u} - \vec{w}) \times \vec{v}\|$ .

- (A)  $10\sqrt{2}$       (B)  $2\sqrt{65}$       (C)  $8\sqrt{6}$       (D)  $12\sqrt{3}$       (E) NOTA

**Please Use the Following Information to Answer Questions 15 to 19:** Let  $\mathcal{G} = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 \\ 0 & 0 & 3 \end{bmatrix}$ .

Define the linear transformation  $\mathcal{G}_\tau : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  by  $\mathcal{G}_\tau(\vec{v}) = \mathcal{G}\vec{v}$ . Let  $\vec{v}_{\mathcal{G}}$  denote the vector  $\vec{v}$  in the  $\mathcal{G}$ -basis, that is,  $\mathcal{G}_\tau(\vec{v}) = \vec{v}_{\mathcal{G}}$ . For example,  $\mathcal{G}_\tau(\langle 5, 5, 5 \rangle) = \langle -1, 7, 15 \rangle = \langle 5, 5, 5 \rangle_{\mathcal{G}}$ .

15. Which of the following is equal to  $\langle -4, -3, 7 \rangle_{\mathcal{G}}$ ?

- (A)  $\langle 0, 5, \frac{7}{3} \rangle$       (B)  $\langle -5, 0, 21 \rangle$       (C)  $\langle 0, -5, 21 \rangle$       (D)  $\langle 5, 0, 21 \rangle$       (E) NOTA

16. Let  $\langle 5, -5, 9 \rangle = \langle \alpha, \beta, \gamma \rangle_{\mathcal{G}}$ . Which of the following is equal to  $\alpha + \beta + \gamma$ ?

- (A)  $-5$       (B)  $-2$       (C)  $0$       (D)  $8$       (E) NOTA

17. Let  $\vec{g}_1 = \langle 17, -5, 0 \rangle_{\mathcal{G}}$ ,  $\vec{g}_2 = \langle -16, 8, 5 \rangle_{\mathcal{G}}$ , and  $\vec{g}_3 = \langle 26, -13, -8 \rangle_{\mathcal{G}}$ . Compute  $|\vec{g}_1 \quad \vec{g}_2 \quad \vec{g}_3|$ , where  $\vec{g}_1$ ,  $\vec{g}_2$ , and  $\vec{g}_3$  are column vectors.

- (A)  $-21$       (B)  $-12$       (C)  $12$       (D)  $21$       (E) NOTA

18. Let  $\mathcal{R}$  be a region in  $\mathbb{R}^3$  defined by a cube with adjacent vertices at  $(-23, 11, -8)$  and  $(-25, 10, -6)$ . Every point in  $\mathcal{R}$  is transformed by  $\mathcal{G}$  to create a region  $\mathcal{R}_{\mathcal{G}}$ . What is the volume of  $\mathcal{R}_{\mathcal{G}}$ ?

- (A)  $9$       (B)  $27$       (C)  $81$       (D)  $243$       (E) NOTA

19. For this question, suppose that  $\mathcal{G} = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}$ , and redefine  $\mathcal{G}_\tau$  and  $\vec{v}_{\mathcal{G}}$  accordingly. Which of the following is equal to  $\langle -4, 3, 2 \rangle_{\mathcal{G}} \times \langle 3, -6, 1 \rangle_{\mathcal{G}}$ ?

- (A)  $\langle 1, 18, 15 \rangle$       (B)  $\langle 4, 17, 15 \rangle$       (C)  $\langle 9, 13, 10 \rangle$       (D)  $\langle 15, 10, 15 \rangle$       (E) NOTA

20. In a very distant future, the Galactic United Infantry (*GUI*) has set up a space highway modeled by the line

$$\mathcal{H} : \langle 3, 0, 3 \rangle + t\langle 5, -1, 7 \rangle.$$

General Jason, a legendary member of *GUI*, finds himself in space at the coordinates  $(-4, 2, -6)$  and needs to get to this highway *pronto*. What is the shortest distance between Jason and the *GUI* highway,  $\mathcal{H}$ ?

- (A)  $\frac{\sqrt{6}}{6}$       (B)  $\frac{\sqrt{6}}{3}$       (C)  $\frac{\sqrt{6}}{2}$       (D)  $\sqrt{6}$       (E) NOTA

21. Compute  $\det \left( \begin{bmatrix} 6 & 1 \\ 11 & -4 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 5 & -9 & -2 \\ 10 & -3 & 0 \end{bmatrix} \right).$

- (A)  $-49$       (B)  $-6$       (C)  $7$       (D)  $36$       (E) NOTA

22. *GUI* Private Hank is moving in space to deliver an incredibly important report to his Captain, Gull. Gull sends Hank his coordinates, so Hank begins moving to the point  $(9, 3, 1)$ . However, that exact day, the “3D-*GUI*-GPS” wasn’t working as intended, so Gull accidentally send Hank his coordinates when reflected across the line

$$\ell : \langle 7, -7, 7 \rangle + t\langle 2, 2, 1 \rangle.$$

Where is *GUI* Captain Gull located?

- (A)  $(13, -9, 17)$       (B)  $(10, -7, 4)$       (C)  $(8, -13, 15)$       (D)  $(2, -6, 8)$       (E) NOTA

23. You’ve certainly encountered geometric series before in some way, but have you ever tried to evaluate a geometric series with matrices? Compute

$$\sum_{g=0}^{\infty} \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ 1 & 0 \end{bmatrix}^g$$

- (A)  $\begin{bmatrix} 2 & -\frac{1}{3} \\ 2 & \frac{1}{6} \end{bmatrix}$       (B)  $\begin{bmatrix} 3 & -\frac{1}{2} \\ 3 & \frac{1}{2} \end{bmatrix}$       (C)  $\begin{bmatrix} 2 & -1 \\ 3 & \frac{1}{2} \end{bmatrix}$       (D)  $\begin{bmatrix} \frac{6}{5} & \frac{6}{7} \\ 1 & 0 \end{bmatrix}$       (E) NOTA

**Please Use the Following Information to Answer Questions 24 to 25:** The Maclaurin series of an algebraic function is a polynomial (possibly of infinite degree) that is equivalent to the function around the point  $x = 0$ . That is, for values bounded around  $x = 0$ ,  $f(x) = \sum_{n=0}^{\infty} \alpha_n x^n$  for  $\alpha_n \in \mathbb{R}$ . For example,  $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$  for all real values of  $x$ . The Maclaurin

series of  $\ln(1+x)$  is  $\sum_{n=1}^{\infty} -\frac{(-x)^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$  for  $-1 < x \leq 1$ .

We can extend this concept to matrices, allowing us to evaluate algebraic functions with matrix arguments! Give it a shot in the following two questions.

24. Compute  $e^{\begin{bmatrix} 4 & 1 \\ 0 & 2 \end{bmatrix}}$ .

- (A)  $\begin{bmatrix} e^4 & e \\ 1 & e^2 \end{bmatrix}$  (B)  $\begin{bmatrix} e^4 & \frac{e^4 - e^2}{2} \\ 0 & e^2 \end{bmatrix}$  (C)  $\begin{bmatrix} e^4 & \frac{e^4 + e^2}{2} \\ 0 & e^2 \end{bmatrix}$  (D)  $\begin{bmatrix} e^4 & e^6 \\ 0 & e^2 \end{bmatrix}$  (E) NOTA

25. Compute  $\ln\left(\begin{bmatrix} 4 & 9 \\ -1 & -2 \end{bmatrix}\right)$ .

- (A)  $\begin{bmatrix} 4 & 9 \\ -1 & -2 \end{bmatrix}$  (B)  $\begin{bmatrix} -3 & 6 \\ -4 & 3 \end{bmatrix}$  (C)  $\begin{bmatrix} 3 & 9 \\ -1 & -3 \end{bmatrix}$  (D)  $\begin{bmatrix} 3 & 8 \\ -2 & -3 \end{bmatrix}$  (E) NOTA

26. *SallesWorld* is a very wacky theme park and the days are extremely random! On any give day, the day can be “Gooley”, “Slimy”, or “Sticky”. However, depending on what happened the day before, the probability of each event changes! Thankfully, *SallesWorld* employee Lance has been keeping track of the days and noted down these probabilities in the following table:

|                               |        | Given that yesterday was ... |               |               |
|-------------------------------|--------|------------------------------|---------------|---------------|
|                               |        | Gooley                       | Slimy         | Sticky        |
| Probability of today being... | Gooley | $\frac{1}{2}$                | $\frac{1}{2}$ | 0             |
|                               | Slimy  | $\frac{1}{2}$                | $\frac{1}{3}$ | $\frac{1}{3}$ |
|                               | Sticky | 0                            | $\frac{1}{6}$ | $\frac{2}{3}$ |

Given that yesterday was “Gooley”, which of the following is closest to the probability that it will be “Gooley” in 777 days?

- (A) 0.33 (B) 0.4 (C) 0.67 (D) 0.75 (E) NOTA

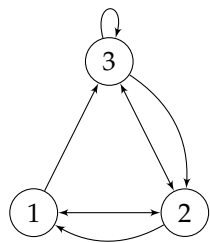
27. Let  $\mathcal{G}$  be a  $3 \times 3$  upper-triangular matrix with distinct non-zero elements that satisfies

$$\mathcal{G}^3 = 2\mathcal{G}^2 + 41\mathcal{G} - 42I$$

If  $\mathcal{G}$  has 6 non-zero elements, find the sum of the digits of  $\det(\mathcal{G}^2)$ .

- (A) 14 (B) 16 (C) 18 (D) 20 (E) NOTA

Please Use the Following Directed Graph to Answer Question 28:



28. Jesse suddenly finds himself at Node 1 and feels compelled to walk around this directed graph. Each minute, starting at minute 1, Jesse walks from one node to another (possibly the same) by taking one of the paths in the graph, following the arrows.

Before Jesse begins moving, “Gooley”, a psychic and mathematician, counted the number of paths that start and end at each combination of nodes after 3 minutes elapse. He noted his findings on the following table:

|               |   | Ending Node |    |   |
|---------------|---|-------------|----|---|
|               |   | 1           | 2  | 3 |
| Starting Node | 1 | 4           | 6  | 6 |
|               | 2 | 8           | 6  | 7 |
|               | 3 | 4           | 10 | 9 |

If, after 3 minutes, Jesse is on Node 2, how many ways are there for him to end up at Node 3 after 7 minutes have elapsed?

- (A) 126
- (B) 176
- (C) 256
- (D) 328
- (E) NOTA

29. At *AaronAlternativeSolutions* (AAS), workers receive a matrix and must take its inverse and print it before shipping it off to *The JiGui Company*. Katie, a worker at AAS, received the following matrix:

$$\mathcal{G} = \begin{bmatrix} 2 & 3 & 1 & 0 \\ 2 & 0 & -2 & 1 \\ -5 & -3 & 2 & -1 \\ 0 & -3 & 0 & 6 \end{bmatrix}.$$

After properly taking the inverse and attempting to print it, Katie realizes that there was a printing error on the last row! Therefore, Katie only printed the following:

$$\mathcal{G}^{-1} = \begin{bmatrix} -4 & -\frac{19}{3} & -\frac{13}{3} & \frac{1}{3} \\ 4 & 6 & 4 & -\frac{1}{3} \\ -3 & -\frac{16}{3} & -\frac{10}{3} & \frac{1}{3} \\ \spadesuit & \clubsuit & \heartsuit & \diamondsuit \end{bmatrix}.$$

Help Katie fix the issue by finding  $\spadesuit + \clubsuit + \heartsuit + \diamondsuit$ .

- (A) 0
- (B)  $\frac{7}{3}$
- (C)  $\frac{14}{3}$
- (D) 7
- (E) NOTA

30. Welcome to the finale of the FAMAT State Convention 2026 Matrices and Vectors test! This was a grueling journey with many highs and lows, but I, the test writer, am incredibly proud that you tackled the questions put before you today!

In order to congratulate this on monumental occasion, for the last time, compute the following determinant, where  $\mathcal{G}$  is the number of times “ $\mathcal{G}$ ” appears on this test:

|               |               |               |                  |                   |                                      |                    |                 |               |    |
|---------------|---------------|---------------|------------------|-------------------|--------------------------------------|--------------------|-----------------|---------------|----|
| 2             | 0             | 0             | 0                | 0                 | 0                                    | 0                  | 0               | 0             | 0  |
| $\mathcal{G}$ | 6             | 0             | 0                | 0                 | 0                                    | 0                  | 0               | 0             | 0  |
| 0             | $\mathcal{G}$ | -1            | 0                | 0                 | 0                                    | 0                  | 0               | 0             | 0  |
| 6             | 0             | $\mathcal{G}$ | $\ln(\pi)$       | 0                 | 0                                    | 0                  | 0               | 0             | 0  |
| 7             | -7            | 0             | $\mathcal{G}$    | $\frac{7}{9}$     | 0                                    | 0                  | 0               | 0             | 0  |
| 1337          | 9             | 10            | 21               | $\mathcal{G}$     | $\log_{\pi}\left(\frac{1}{2}\right)$ | 0                  | 0               | 0             | 0  |
| $\pi$         | $2\pi$        | $3\pi$        | $4\pi$           | 0                 | $\mathcal{G}$                        | $\frac{1}{\ln(2)}$ | 0               | 0             | 0  |
| $\cos(1)$     | $\sin(1)$     | $\tan(1)$     | $\sec(1)$        | $\csc(1)$         | $\cot(1)$                            | $\mathcal{G}$      | $-\frac{1}{14}$ | 0             | 0  |
| $-\sin(1)$    | $\cos(1)$     | $\sec^2(1)$   | $\sec(1)\tan(1)$ | $-\csc(1)\cot(1)$ | $-\csc^2(1)$                         | 0                  | $\mathcal{G}$   | 1013          | 0  |
| 1             | 2             | 3             | 4                | 5                 | 6                                    | 7                  | 8               | $\mathcal{G}$ | -6 |

- (A) 1013
- (B) 2026
- (C) 3039
- (D) 4052
- (E) NOTA